

Factoring Perfect Square Trinomials and More

In this lesson you will learn many new and cool concepts about perfect square trinomials. Before you begin this wonderful journey, you will need to know how to square a binomial. What is a **binomial** you ask? Well, it is simply a polynomial with **two terms**. (By the way, a **trinomial** is a polynomial with **three terms**!) Let's look at some problems below.

Partner, Group, or Independent practice (Hopefully you pick a partner or group! But if you are tired and grumpy because you didn't get a good night sleep, you may work by yourself.)

1. Write **three** equivalent expressions for 5^2 .

Answer will vary: $(9 - 4)^2$, $(1 + 4)^2$

Students can check **ALL** of the answers on this page with the PowerPoint presentation as well!

Hint: To get you started, here is one equivalent expression: $5^2 = (3 + 2)^2$. Now you need to come up with two more! Your welcome!!

2. Write **three** equivalent expressions for 12^2 . (If you are stuck, don't be shy and look at the hint above!)

Answer will vary: $(11 + 1)^2$, $(14 - 2)^2$, $(20 - 8)^2$

Simplify the following expressions for #3–6

3. 10^2 **100**

4. $(3 + 7)^2$ **100**

5. $(11 - 1)^2$ **100**

6. $(-14 + 4)^2$
100

You will now watch a **PowerPoint presentation** with your class to guide you through problems A and B. *Please copy down the steps as you go!*

How to Use use a rectangular diagram to model binomial numerical expressions, Rock On!

A. $(3 + 7)^2$ ←

This problem is the same as problem 4. It is just simplified in a different way!

Please have students watch the step by step animated PowerPoint for this problem!

	3	7
3	9	21
7	21	49

$$9 + 21 + 21 + 49 = 100$$

B. $(10 - 3)^2$ ←

Note: Even if lengths aren't negative, the rectangular model still works with subtraction.

Please have students watch the step by step animated PowerPoint for this problem!

	10	-3
10	100	-30
-3	-30	9

$$100 + (-30) + (-30) + 9 = 49$$

Please continue to the next page.

For problems #7-12, use what you have learned to **make a rectangular diagram to model the following expressions!**

7. $(1 + 3)^2$

16

8. $(20 + 2)^2$

484

Students can check **ALL** of the answers on this page with the PowerPoint presentation as well!

9. $(7 + 30)^2$

1,369

10. $(15 - 1)^2$

196

You will continue to watch the **PowerPoint presentation** with your class to guide you through problems C and D. *Please copy down the steps as you go!*

C. $(x + 4)^2$ ←

How to use a rectangular diagram to expand a binomial that contains a variable.

Please have students watch the step by step animated PowerPoint for this problem!

	x	4
x	x^2	$4x$
4	$4x$	16

$$x^2 + 4x + 4x + 16 = x^2 + 8x + 16$$

D. $(x - 7)^2$ ←

How to use a rectangular diagram to expand a binomial that contains a variable.

Please have students watch the step by step animated PowerPoint for this problem!

	x	-7
x	x^2	$-7x$
-7	$-7x$	49

$$x^2 + (-7x) + (-7x) + 49 = x^2 - 14x + 49$$

Partner, Group, or Independent Practice: For problems #11–16 use a rectangular diagram to expand the following binomials!

11. $(x + 5)^2$

$$x^2 + 10x + 25$$

12. $(x - 9)^2$

$$x^2 - 18x + 81$$

Students can check the answers to **problems #11 and #12** with the PowerPoint presentation as well!

Continue to use rectangular models to expand the following binomials.

13. $(x + 3)^2$

$$x^2 + 6x + 9$$

14. $(x - 15)^2$

$$x^2 - 30x + 225$$

Students can check **ALL** of the answers
on this page with the PowerPoint
presentation as well!

15. $(x - 30)^2$

$$x^2 - 60x + 900$$

16. $(x + 16)^2$

$$x^2 + 32x + 256$$

You will continue to watch the **PowerPoint presentation** with your class to guide you through problems E and F. *Please copy down the steps as you go.* Math Rocks!

How to make rectangular diagrams for trinomials and write them in the form $(x + h)^2$ or $(x - h)^2$.

E. $x^2 + 6x + 9$
 $(x + 3)^2$

Please have students watch the step by step animated PowerPoint for this problem!

F. $x^2 - 8x + 16$
 $(x - 4)^2$

Please have students watch the step by step animated PowerPoint for this problem!

Partner, Group, or Independent Practice: For problems #17–20 make rectangular diagrams for the following trinomials. Write your final “answer” in the form $(x + h)^2$ or $(x - h)^2$.

17. $x^2 + 10x + 25$
 $(x + 5)^2$

18. $x^2 + 20x + 100$

$(x + 10)^2$

Students can check the answers to **problems #17 and #18** with the PowerPoint presentation as well!

Continue to make rectangular diagrams for the following trinomials. Write your final “answer” in the form $(x + h)^2$ or $(x - h)^2$.

19. $x^2 - 14x + 49$

$(x - 7)^2$

20. $x^2 - 26x + 169$

$(x - 13)^2$

Students can check **ALL** of the answers on this page with the PowerPoint presentation as well!

21. Use a magnifying glass to look closely at the original problems in #17–20. You should notice that all of these expressions are trinomials. They are also special trinomials called **perfect square trinomials** because there is a unique relationship between the middle term and the last term of the expression. **Can you identify this relationship?** (Hint: What can you do to the coefficient of the middle term to get to the last term?)

Another Hint:

$$x^2 + 10x + 25$$

What operations can you perform on the 10 to “turn it into” a 25?

Write the relationship between the middle and last term in the box below!

If you divide the coefficient of the middle term by 2, then square it, it equals the last term!

****Note the coefficient of x^2 is 1 in these situations!**

You will continue to watch the **PowerPoint presentation** with your class to guide you through problems G, H, and I. *Sad news! You are more than half way through this lesson!*

G. $x^2 + 12x + 36$

How to identify if a trinomial is a perfect square trinomial. And if it is, how to write it in the form $(x + h)^2$ or $(x - h)^2$

Please have students watch the step by step animated PowerPoint for this problem!

Yes: $(x + 6)^2$

H. $x^2 - 30x + 225$

How to identify if a trinomial is a perfect square trinomial. And if it is, how to write it in the form $(x + h)^2$ or $(x - h)^2$

Please have students watch the step by step animated PowerPoint for this problem!

Yes: $(x - 15)^2$

I. $x^2 - 4x + 2$

How to identify if a trinomial is a perfect square trinomial. And if it is, how to write it in the form $(x + h)^2$ or $(x - h)^2$

Please have students watch the step by step animated PowerPoint for this problem!

This is not a perfect square trinomial.

For problems #22–27, Identify if the following trinomials are perfect square trinomials. If it is a perfect square write it in the form $(x + h)^2$ or $(x - h)^2$

22. $x^2 - 12x + 36$

Yes, This **is** a perfect square trinomial.

$$(x - 6)^2$$

23. $x^2 + 16x + 64$

Yes, This **is** a perfect square trinomial.

$$(x + 8)^2$$

Students can check **ALL** of the answers on this page with the PowerPoint presentation as well!

24. $x^2 - 18x + 18$

No, this is not a perfect square trinomial.

25. $x^2 - 20x - 100$

No, this is not a perfect square trinomial.

26. $x^2 + 14x - 49$

No, this is not a perfect square trinomial.

27. $x^2 - 3x + \frac{9}{4}$

Please oh please, don't start off this problem saying I hate fractions.

Yes, This **is** a perfect square trinomial.

$$\left(x - \frac{3}{2}\right)^2$$

You will continue to watch the **PowerPoint presentation** with your class to guide you through Problem J. Please copy down the steps as you go!

J.

$$x^2 - 14x + 49 = 4$$



How to solve an equation that contains a perfect square trinomial.

Please have students watch the step by step animated PowerPoint for this problem!

$$x = 9 \quad \text{or} \quad x = 5$$

Please continue to the last page of this packet.

Solve the following equations for x . (Note: All of these equations contain perfect square trinomials.) **Also, you're encouraged to check your solutions.** If you're not sure how to check your solutions, you can always ask your benign partner, group or teacher!!

28. $x^2 + 12x + 36 = 49$

$x = 1$ or $x = -13$

29. $x^2 + 4x + 4 = 9$

$x = 1$ or $x = -5$

Students can check **ALL** of the answers on this page with the PowerPoint presentation as well!

30. $x^2 - 100x + 2500 = 1600$

$x = 90$ or $x = 10$

31. $x^2 - x + \frac{1}{4} = 289$

$x = 17.5$ or $x = -16.5$

Hint: $-x$ is equivalent to $-1x$. **Oh and remember, you love fractions!**